BullSO 2026 Invitational

Crime Busters B

Bull Science Olympiad at the University of South Florida

Saturday, January 24th, 2026

Team Name:_____

Participant Names:_____

Written Test Score:_____**/200** **Rank:**_____

Section A: The Crime

Three days after the devastating Louvre Heist, the director decided to have a grand reopening of the museum. The staff excitedly prepared the exhibit, making sure everything was in place. As the newly refurbished exhibit was revealed, people began to admire the sights. They enjoyed baked goods, such as croissants, eclairs, baguettes, and macarons, which were all provided by the Louvre bakery. The room was adorned with peonies, roses, and lilies, which were provided by the Louvre's gardener. All in all, the event was turning out to be a success, or so they thought.

After some time, the director heard a crash coming from a closed off room. Leaving the main event, he ran to the room. He frantically looked around the room, only to find…

A crown appearing to be floating in midair, surrounded by shattered glass!

Devastated, the director called security to search for the culprit, but it was too late. The crown cushion was nowhere to be seen!

After searching the scene, investigators found numerous pieces of evidence including 4 powders, 1 mixture, 2 liquids, 1 metal, 2 plastics, 2 fibers, 3 hairs, a note, fingerprints, and shattered glass. The following suspects were in attendance at the reopening and were confirmed to be there around the time it happened. The director has hired you, the forensic detective, to analyze the evidence and determine the culprit. Good luck! 😉

Suspect #1: Napoleon Croissant

Napoleon Croissant snuck into the reopening event despite not being invited and having a restraining order filed against him by the director for stalking. He believes that he is Napoleon Bonaparte reincarnated, and therefore believes that the crown cushion is rightfully his. During the event, he was drinking lemonade. Upon being questioned by the investigators, he revealed that he is taking Alka-Seltzer for heartburn and the investigators noted that he seems to be delusional. He has black hair.

Suspect #2: Camille Croquembouche

Camille Croquembouche is the head baker of the bakery inside the Louvre Museum. She has been working there for over 2 decades and has started to feel a little unappreciated. Her coworkers have heard her complaining about the workload. The night before the reopening, she had an argument with the director about how her cafe deserves a renovation just like the museum. On the day of the reopening, she provided catering for the guests. She was wearing an apron made of linen, a plant fiber, and a hairnet covering her blonde hair.

Suspect #3: Pascal Paintpot

Pascal Paintpot, who the investigators first thought was his famous brother Pedro Paintpot, was painting a picture of the room. His brother, who had a time-conflicting event, told him to go in his place since his twin brother was losing his reputation as a painter. Pascal thought this job for the Louvre would save his reputation, so he agreed. He dyed his hair red to match his brother's, wore a shirt made of polyester, a synthetic fiber, and confidently attended the event. Investigators noticed that he had a bandage around his arm, which he said was covering a cut from an old accident.

Suspect #4: Frank Normal

Frank Normal, a true redhead, was invited to the reopening by his friend, the director. He was planning on using this opportunity to get close to Francine Fleur, the gardener, who he's had a crush on for two years. When the cushion under the crown was noticed to be gone, he was wearing a shirt made of cotton, a plant fiber, and was spotted with Francine. When being questioned by the investigators, he answered each question nonchalantly, almost as if he was hiding something.

Suspect #5: Francine Fleur

Francine Fleur is the gardener of the Louvre Museum, maintaining the Tuileries Garden and taking care of her pet squirrel, Peanut. She has been working there for 4 years. She has black hair. On the day of the reopening, she provided flowers for the event. When the cushion under the crown was noticed to be gone, she was wearing a sweater made of wool, an animal fiber, and was spotted with Frank Normal. When being questioned by the investigators, she revealed that she had been in a relationship with the museum director for 3 years before the director ended their relationship. Investigators noted that Francine sounded bitter about the breakup.

Section B: Qualitative Analysis (80 pts)

Powders	Name (2 pts)	Method of Identification (3 pts)	Who does this implicate and why?
#1	Flour	Insoluble in water Looks like flour.	Implicates Camille because flour is a common baking ingredient and she is a baker.
#2	Sugar	Soluble in water	Implicates Camille because sugar is a common baking ingredient and she is a baker.
#3	Cornstarch	Iodine test, turns blue-black	Implicates Camille because cornstarch is a common baking ingredient and she is a baker.
#4	Sodium Hydrogen Carbonate (Baking Soda)	Reacts rapidly with HCl, fizzy bubbles form	Implicates Camille because $NaHCO_3$ is a common baking ingredient and she is a baker.

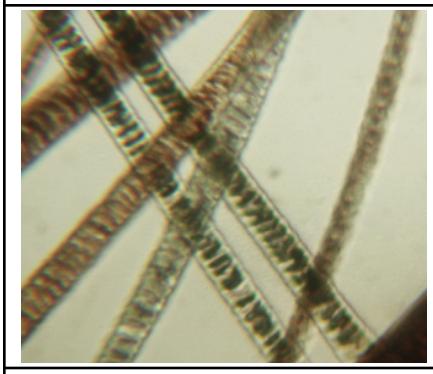
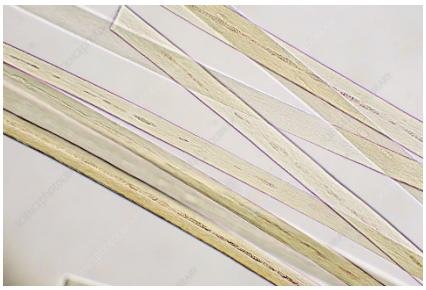
Mixture	Name (2 pts)	Method of Identification (3 pts)	Who does this implicate and why?
Mixture of 2 materials	Sand and powdered Alka-Seltzer	Color of sand (looks like sand) Fizzy bubbles rapidly form when Alka-Seltzer reacts with water	Francine because sand is often used in gardens and she is a gardener. Napoleon because he has heartburn issues.

Liquids	Name (2 pts)	Method of Identification (3 pts)	Who does this implicate and why?
A	Water	Neutral (pH = 7)	Implicates no one
B	Lemon Juice	Acidic, smells like lemon	Implicates Napoleon because he was drinking lemonade during the event.

Metal	Name (2 pts)	Method of Identification (3 pts)	Who does this implicate and why?
M	Aluminum	Not magnetic, silver, malleable	Implicates no one

Section C: Polymers (34 pts)

Fibers	Fiber Type (plant, animal, synthetic) (1 pt)	Who does this implicate and why? (3 pts)
#1	Plant	Implicates Camille because she wore a linen apron.
#2	Animal	Implicates Francine because she wore a wool sweater.

Hair	Type of Hair (1 pt)	Who does this implicate and why? (3 pts)
	Cat	Implicates Francine because she has a pet cat named Peanut
	Human	Implicates Camille because she has blonde hair.
	Human	Implicates Pascal and Frank have red hair.

Plastic Description	Full Name & Abbreviation (2 pts)	Who does this implicate and why? (3 pts)
• Burn test: blue with yellow tip, continues to burn when the source of the fire is removed • Floats in water, floats in alcohol, floats in oil	PP, polypropylene	Implicates Napoleon because PP is used in prescription bottles and he takes medication, which is stored in prescription bottles.
• Burn test: blue with yellow tip, smells like candle wax or paraffin • Floats in water, sinks in alcohol, floats in oil	HDPE, high density polyethylene	Implicates Francine because HDPE is used in flowerpots and she is a gardener and would likely use flowerpots.

Section D: Chromatography (15 pts)

The following note was found at the crime scene:



Directions: Please attach each of the chromatography strips to this page. Then, calculate the retention factor (Rf) for each pigment on the chromatography strips. Finally, identify which suspect's pen matches that of the pen found at the crime scene.

Section E: Crime Scene Physical Evidence (18 pts)

Fingerprint Analysis

Suspect Name	Fingerprint	Type of Pattern (1 pt)
Napolean Croissant		Double loop whorl
Camille Croquembouche		Arch
Pascal Paintpot		Ulnar loop

Suspect Name	Fingerprint	Type of Pattern (1 pt)
Frank Normal		Tented arch
Francine Fleur	 tree	Tented arch
Crime Scene		Arch

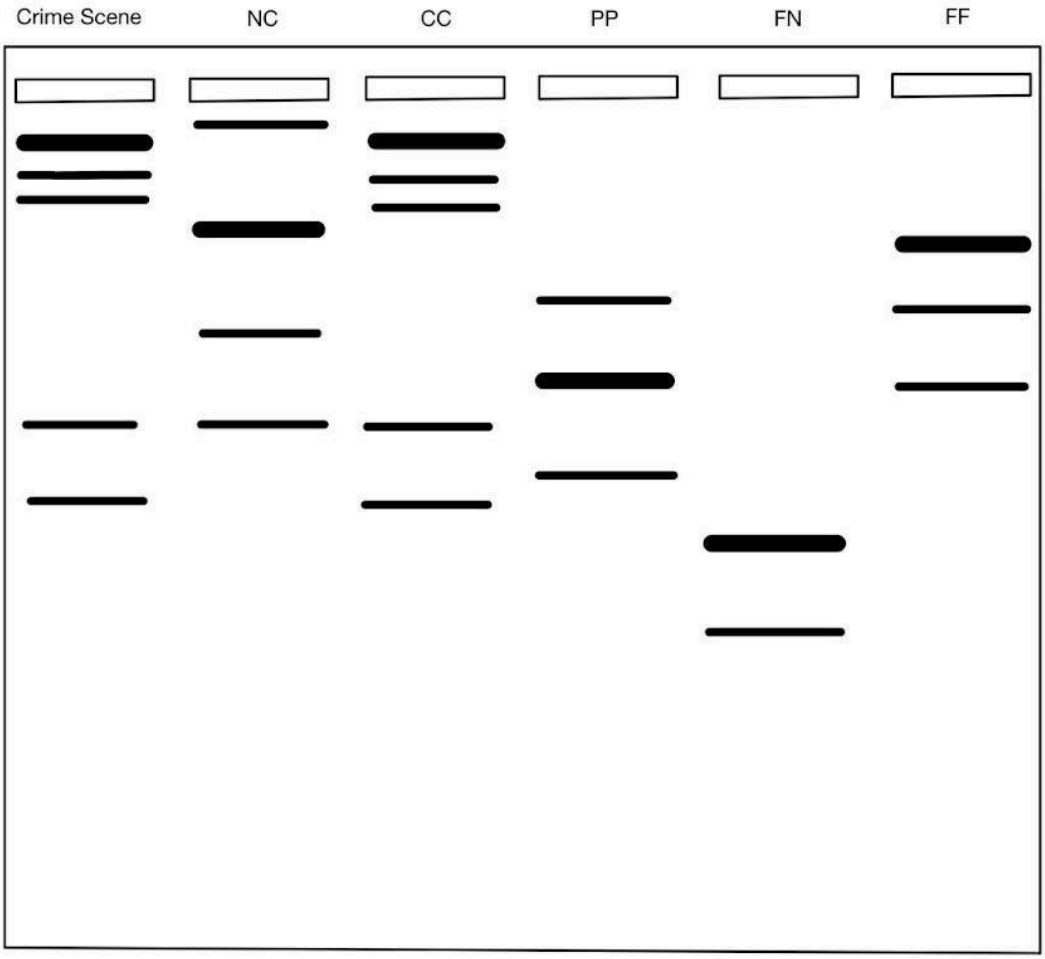
Who does the crime scene fingerprint implicate and why? (3 points)

Camille because both her and the crime scene have a plain arch pattern and the same minutiae.

Why do all of our fingers have a unique fingerprint? (3 pts)

The identifying details (minutiae) arise from random events and environmental factors in the womb, like amniotic fluid pressure, fetal movement, and nerve growth, making each finger's ridge formation a chaotic, one-of-a-kind process.

DNA Analysis



Based on the gel electrophoresis, who does the crime scene DNA sample implicate and why? (3 points)

Camille because her DNA bands match up completely with the one at the crime scene.

What does PCR stand for? (3 points)

Polymerase Chain Reaction

Section F: Analysis of Crime Scene (55 pts)

Using evidence you have been presented with, write an analysis of the crime scene. Make sure to mention all suspects, discuss evidence that links them to the crime (and how), and any evidence that implicates no one. Discuss which suspect is the culprit and why you believe so, as well as why you believe the others did not commit the crime.

Point Breakdown:

- Identification of Camille as the culprit (10 pts)
- Mentions of Camille's motive (4 pts)
- Evidence: (3 pts each except powders which are 4 pts)
    - Powders (baking soda, cornstarch, sugar, flour) implicate Camille (baker)
    - Plant fiber and blonde hair found at crime scene (Camille has blonde hair)
    - Fingerprints (Crime scene and Camille's match)
    - DNA (Crime scene DNA and Camille's DNA matches)
- Explanation of evidence (cannot just list evidence with no explanation of correlation to the suspect) (12 pts, 3 pts for each evidence)
- Why it isn't other suspects (12 pts, 3 pts for each suspect)
- Doesn't implicate anyone (4 pts)
    - Water
    - Aluminum

BullSO, University of South Florida - Repaired practice edition - Answer key

Division B | Crime Busters | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

repair-case. (4 points) Authored written substitute for unavailable source material: Why is plastic evidence class evidence?

Unit: Recyclable plastics.
Worked-case data: A plastic fragment sinks in water, floats in a 1.10 g/cm^3 salt solution, and burns with a sooty flame. Identify it.
Separate investigation observations:
Water: Sinks.
1.10 g/cm^3 solution: Sinks.
1.25 g/cm^3 solution: Floats.
Appearance: Very clear, rigid, and shatter-resistant.

answer: The same polymer is made into many products, so a match shows consistency with a source type, not one specific object. (4 points).